Final Report

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Part IIA Project: *Neural Control with Adaptive State Estimation* (GG4)
June 11, 2026

Introduction

We are given a Brain and a Brain–Machine Interface (BMI), which controls an arm. We are tasked with controlling the position of the hand at the end of the arm. The Brain takes $n_u = 2$ inputs and produces $n_y = 16$ outputs; we assume it is a Linear Gaussian State–Space Model (LGSSM). The BMI takes the 16-dimensional observations from the Brain and applies a fixed linear decoder to map them to 2 latent states: x_1, the muscle selection rhythm; and x_2, power rhythm. The BMI stores a rolling history of x_1, which is passed through 4 fixed band–pass filters which measure how strongly x_1 oscillates at that muscle's defined excitation frequency. A history of x_2 is also stored in state, and the current value is compared to a half– and full–period in the past: if strong oscillation is found, evidence accumulates; otherwise, evidence decays, producing a smooth power state. Muscle activation is the product of activation and the selection vector. Each muscle produces force proportional to its activation, and for each joint, the net force is the sum of signed muscle forces. Positive forces represent flexion; negative forces represent extension. A dry-friction threshold must be overcome, and once this is overcome, the remaining net force (once subtracting the dry friction) is divided by wet friction representing viscous resistance. Joint angles are integrated, and 2-link arm kinematics give the hand position.

Rules of engagement

The holistic view of this project is to be able to reason and interpret a controller which takes observations from a brain (e.g. via electrodes on the scalp) to move a prosthetic arm. Critically, the brain is a black box and its model "parameters" cannot be directly observed. In the spirit of this view, we propose specific rules of engagement surrounding our analysis. We do not attempt to decompile the provided Brain executables to derive the system matrices. We do not average over multiple Brain seeds, removing noise to derive system matrices, we reserve this for evaluation purposes only to demonstrate robustness against noise. Where we do run experiments that are not physically realisable, we do this to demonstrate the direction of our analysis, and we replicate results in empirically valid ways alongside. We do not directly use the weights of the BMI in our control efforts: we allow ourselves to read their values for experimental design, but any results we must prove empirically before use. We only allow the use of the 16-dimensional observations from the Brain and the hand position given by the BMI for closed-loop control; we do not rely on the internal state of the BMI (e.g., arm angles or latent state variables) for feedback. We also do not expect or attempt to control the elbow joint past 180° to mimic the capabilities of an anatomically accurate arm. We motivate these constraints as to achieve resilience and independence to both: inter-individual variation in brains (and therefore system matrices); and manufacturers of prosthetic arms (and therefore BMI decoding, state, and arm kinematics defined by model weights).

System identification

As we assume the Brain is a LGSSM, we explore system identification as a means to understand the system matrices defining the model:

$$\begin{aligned} \underline{x}_{t+1} &= \underline{A}\,\underline{x}_t + \underline{B}\,\underline{u}_t + \underline{w}_t, \qquad & \underline{w}_t &\sim \mathcal{N}(\underline{0}, \underline{Q}) \\ \underline{y}_t &= \underline{C}\,\underline{x}_t + \underline{o}_t, & \underline{o}_t &\sim \mathcal{N}(\underline{0}, \underline{R}) \end{aligned}$$

Identification evaluation

To evaluate our estimates, we have two strategies: the reconstruction of observations and the reconstruction of frequency response. To measure the reconstruction of observations, we use the coefficient of determination defined with the Frobenius norm:

$$R^2 = 1 - \frac{\left\| Y - \widehat{Y} \right\|_F^2}{\left\| Y - \overline{Y} \right\|_F^2}$$

This form is chosen for its effectiveness at aggregating errors across all time points and all observation dimensions. We weight this up against its disadvantages, which include ignorance of observation covariance and temporal correlations. To measure the reconstruction of frequency response, we inspect a frequency map derived analytically from estimates of system matrices, and compare characteristics with empirical results from a a single run of the Brain. We find the transfer function of the LGSSM, dependent on only the deterministic components (matrices $\underline{A}$, $\underline{B}$, and $\underline{C}$):

$$\underline{H}(z) = \underline{C}\left(z\underline{I} - \underline{A}\right)^{-1}\underline{B}$$

We define an estimate to reconstruct the frequency response well if it can minimise the error between analytic and empirical values of $|H(e^{j\omega})|$ across all neurones and frequencies $[0, \pi]$. Given the emphasis on rhythm in the BMI we weigh frequency response reconstruction more than observation reconstruction if a compromise is necessary.

Identification methods

We run two `Brain` instances with and without input to isolate and subtract the noise. We can then calculate the exact Markov parameters, construct Hankel matrices, and form a singular-value spectrum. With this analysis we find σ_i/σ_1 fall after $i = 6$, indicating the true latent dimension number to be $n_x = 6$ (Figure 1). However, this method is not physically realisable: you cannot run the same physiological brain twice with the same random state. This would mean rewinding a brain, re-run a trial without a stimulus, and subtract: noise is truly stochastic outside of a simulation. A physically realisable single-run Ordinary Least Squares (OLS) regression estimate uncovers a noise floor that obscures modes 3–6. Balancing likelihood and model complexity with Bayesian Information Criterion (BIC) model selection, we identify $n_x = 2 \neq 6$ (Figure 1).

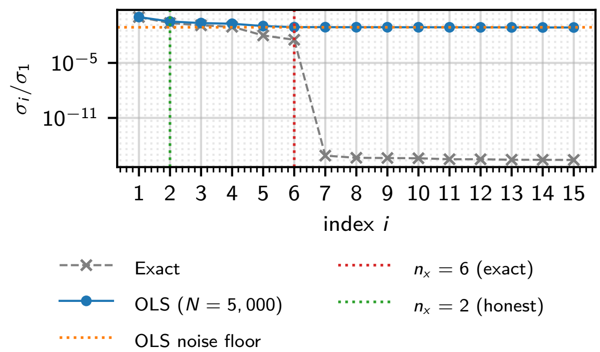

Figure 1: Normalised Hankel singular values σ_i/σ_1 for the 40×60 block Hankel matrix ($n_u = 2$, $n_y = 16$). The exact spectrum is obtained by same-seed impulse subtraction, which is not physically realisable, and serves as a theoretical reference for the true system order. Modes 3–6 fall below the OLS noise floor of a 5000-step physically realisable recording; BIC model selection on the OLS estimates recovers $n_x = 2$. Source

We plot a heatmap of $|H(e^{j\omega})|$ and find that Canonical Variate Analysis (CVA) and Expectation-Maximisation (EM) to reconstruct u_1 well, but not u_2 over all n_x candidates $\{2, 4, 6\}$ (Figure 4, N.B. only $n_x = 4$ plotted). We turn to an estimation method that uses the Eigensystem Realisation Algorithm (ERA) and EM that utilises OLS-estimated Markov parameters and find similar results for $n_x = 2$: good reconstruction of u_1 but poor for u_2. However, in contrast to CVA+EM, reconstruction of the u_2 frequency response improved with $n_x = 4$ and $n_x = 6$ (Figure 4). More generally, we observe that the `Brain` acts as a low-pass filter. The observation reconstruction quality remained similar between CVA+EM and ERA+EM (Figure 2).

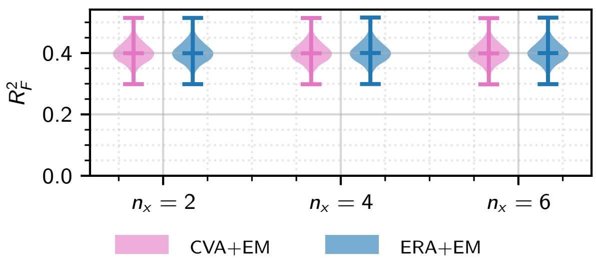

Figure 2: Frobenius R_F^2 of one-step-ahead Kalman predictions for CVA+EM and ERA+EM at $n_x \in \{2, 4, 6\}$, across $N = 1000$ held-out trials of $T = 2000$ steps (first 100 steps excluded as Kalman transient). Median R_F^2: $n_x = 2$: CVA+EM $= 0.399$, ERA+EM $= 0.399$; $n_x = 4$: CVA+EM $= 0.399$, ERA+EM $= 0.400$; $n_x = 6$: CVA+EM $= 0.399$, ERA+EM $= 0.400$. Source

As there was not a huge difference between $n_x = 4$ and $n_x = 6$, we select ERA+EM and $n_x = 4$, which yields system matrices visualised in Figure 3.

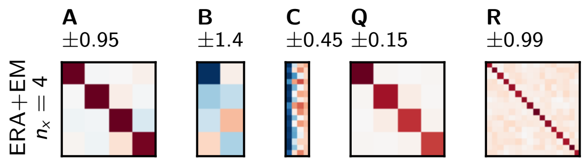

Figure 3: Identified system matrices for ERA+EM ($n_x = 4$). Red indicates positive, blue negative; colour scales are symmetric about zero. Source

Closed-loop spectral control of brain observations

We test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population mean activity. We motivate the following evaluation framework and method with the emphasis on controlling rhythms in the `Brain` observations for influence over BMI latent variables and the arm downstream.

Spectral control evaluation

We use the following metrics to evaluate each controller: spectral Root Mean Square (RMS) error, which measures amplitude deviation at target frequencies; the maximum amplitude error, the worst-case per-component deviation; and the off-target power ratio, the mean spectral amplitude at non-target frequencies, relative to mean target amplitude.

Spectral control method

We offer 3 candidates for control strategies: Linear Quadratic Gaussian (LQG) to respect the `Brain`'s LGSSM structure; Linear Quadratic Integral (LQI) to

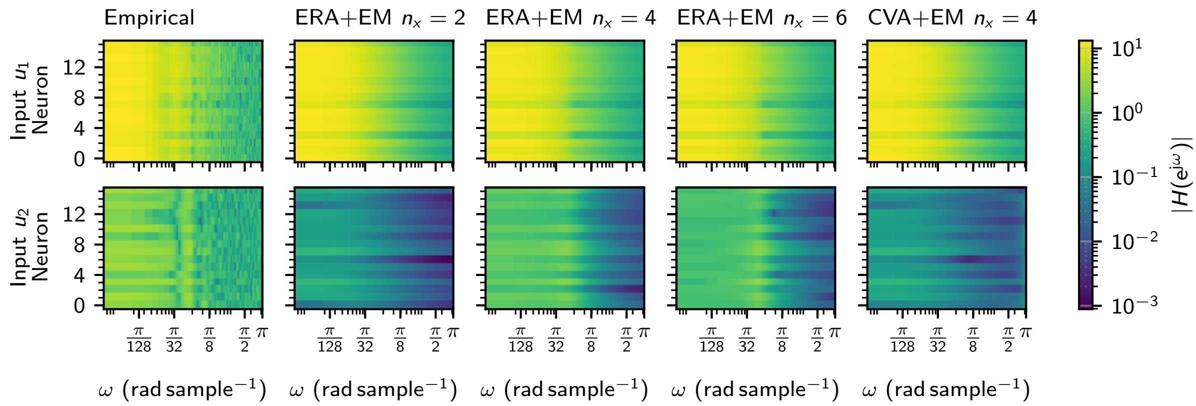

Figure 4: Per-neuron frequency selectivity for inputs u_1 (top row) and u_2 (bottom row). Each panel shows $|H(e^{j\omega})|$ for all $n_y = 16$ neurons (vertical axis) across $\omega \in [0, \pi]$ rad sample^{-1} (horizontal axis), on a shared logarithmic colour scale. Leftmost column: empirical response (honest OLS estimate from a single 5000-step recording, DFT zero-padded to 512 points). Columns 2–4: analytic ERA+EM transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ for $n_x \in \{2, 4, 6\}$ respectively, showing the latent dimension sweep. Rightmost column: CVA+EM at $n_x = 4$, for estimator comparison at the chosen order. Source

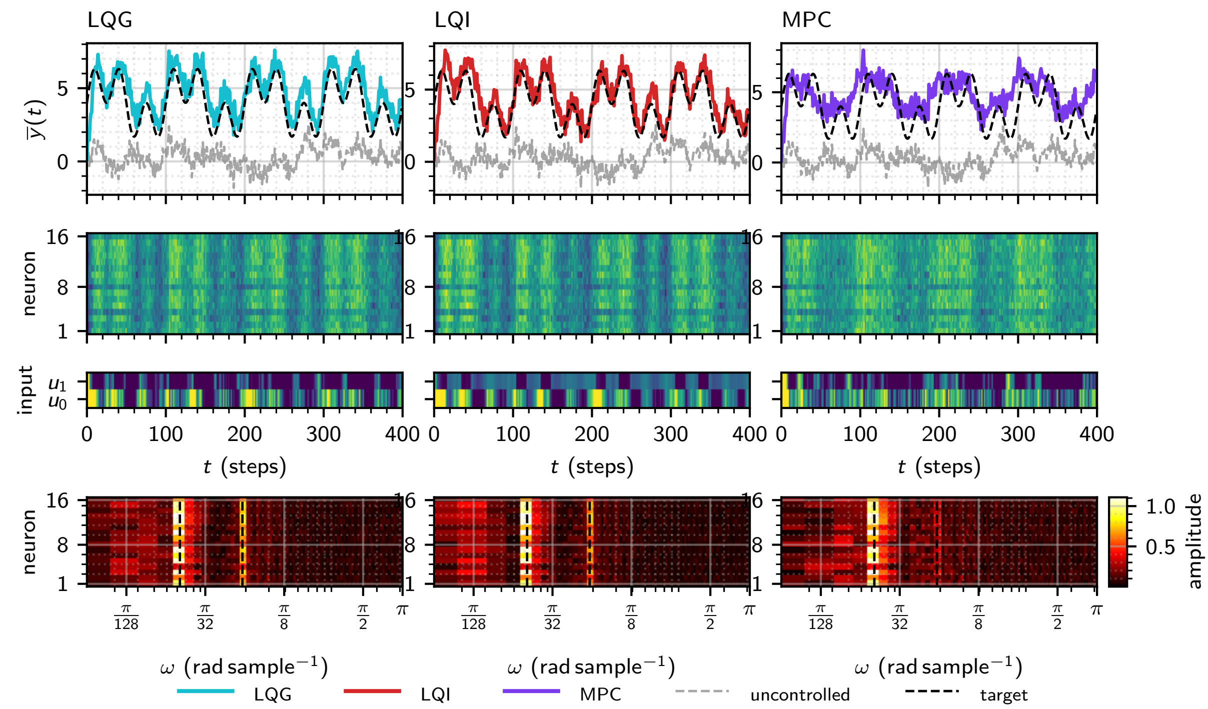

Figure 5: Closed-loop spectral control using the ERA+EM $n_x = 4$ model ($n_x = 4$), controller hyperparameters tuned by random search (8 trials each). The target is a sum of two sinusoids at $f \in \{0.01, 0.03\}$ cycles per sample with amplitudes $A \in \{1.50, 1.50\}$, centred on a physical mean of 4. Each column shows one controller (LQG, LQI, MPC) with a per-neuron tracking cost ($\lambda_{\text{var}} = 1$). Row 1: physical population mean $\overline{y}(t)$, uncontrolled baseline (grey dashed), and target reference (black dashed). Row 2: per-neuron observations $y(t)$ as a neuron $\times$ time heatmap. Row 3: applied inputs $u(t)$. Row 4: per-neuron amplitude spectral density (neuron $\times$ angular frequency ω, log scale); target frequencies shown as red dashed lines. Source

correct steady-state error; and Model Predictive Control (MPC) to optimise for and appreciate constraints on $u_i \in [0, 1]$.

Spectral control results

Inspecting the raw population mean traces, we find LQG overestimates the population mean with a steady-state error, which is removed effectively by LQI. MPC fails to reconstruct the higher frequency components. We select LQI due to its lower control effort compared to LQG and better reconstruction performance over MPC.

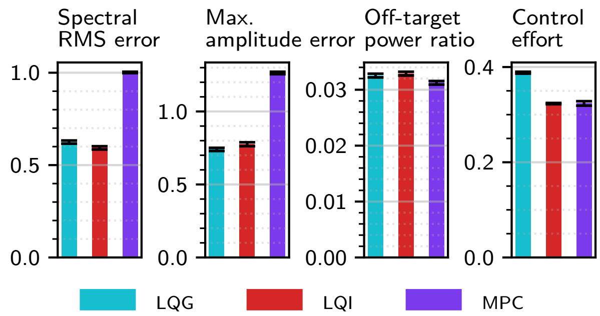

Figure 6: Spectral tracking metrics for LQG, LQI, and MPC controllers using the ERA+EM $n_x = 4$ model ($n_x = 4$). Each controller was tuned by random search (8 trials, seed 1) and evaluated over $N = 20$ independent brain seeds. The target was a sum of sinusoids at $f \in \{0.01, 0.03\}$ cycles per sample with amplitudes $A \in \{1.50, 1.50\}$, centred on a physical mean of 4. Bars show the mean; error bars show the standard error of the mean. Left: spectral RMS error (RMS amplitude deviation at target frequencies). Centre: maximum amplitude error (worst-case per-component deviation). Right: off-target power ratio (mean spectral amplitude at non-target frequencies relative to mean target amplitude). Source

1 Hand control

The mechanism of the BMI of muscle selection and a shared power state means that simultaneous control of different muscles is unlikely, as there is no granular control to direct different muscles at different power at the same time. This constrains our control strategy to be a sequential one, directing one muscle at a time.

Evaluating hand control

We aim to minimise the final distance to target, the Euclidean distance between the hand and the target at the end of a trial. We define a success threshold at 10. We aim to maximise the reach rate, the fraction of trials that reach within this success threshold. This metric captures reliability across diverse targets in the reachable area: a controller with low mean final distance distance but occasional failures is not practically useful for a prosthetic. We minimise the time to threshold, the number of steps until the hand first enters the distance success threshold: a useful prosthetic should converge in reasonable time to provide utility. We maximise the sector fraction, the proportion of time the hand spends in the optimal sector: an angular wedge containing the target and the start position. We minimise the rotational distance, the total angle swept by the hand trajectory over the trial. These two metrics penalise inefficient trajectories (e.g. looping behaviours), which would be inconvenient and unpredictable for a prosthetic. We minimise the control effort, the sum of square brain inputs, in an effort to make inputs more interpretable and efficient.

Muscle selection

We sweep frequencies for both inputs and measure shoulder and elbow angles, determined by inverse kinematics to find the specific frequencies that excite specific muscles. We empirically recover the frequency bands defined in the BMI weights using a single seed, shown in Table 1 and Figure 7.

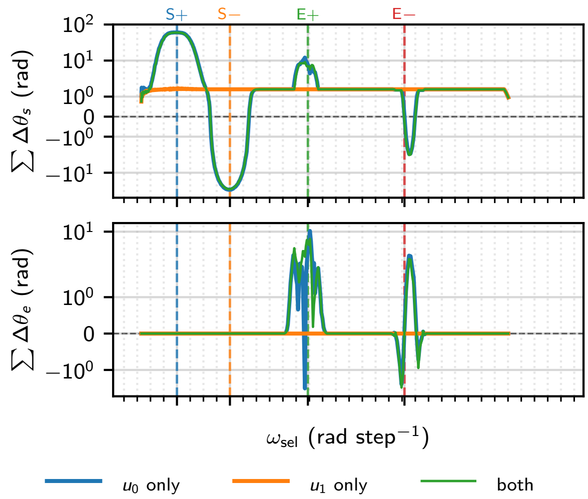

Figure 7: Muscle-selection frequency sweep. At each of 512 logarithmically spaced angular frequencies $\omega_{\text{sel}} \in [0.188, 2.639]$ rad step^{-1}, open-loop sinusoidal drive (amplitude $A = 0.45$, centred at 0.5) is applied to u_0 or u_1 independently. The arm is observed via hand position and inverse kinematics. Top: signed cumulative shoulder displacement $\sum \Delta\theta_s$; bottom: cumulative elbow displacement $\sum \Delta\theta_e$, each summed over the post-warmup window of 900 steps (symlog y-axis, linear threshold ± 1 rad); curves smoothed with a 5-sample moving average. Coloured vertical lines mark the four empirically identified band frequencies (S+ at $\omega^* = 0.426$ rad step^{-1}, S- at $\omega^* = 0.780$ rad step^{-1}, E+ at $\omega^* = 1.301$ rad step^{-1}, E- at $\omega^* = 1.946$ rad step^{-1}), identified from the open-loop u_0 response by peak displacement magnitude (S±) and peak elbow selectivity $|\Delta\theta_e|/(|\Delta\theta_e| + |\Delta\theta_s|)$ (E±). Source

Muscle	ω^*_{true} (rad step^{-1})	ω^*_{rec} (rad step^{-1})	$\Delta\omega$ (rad step^{-1})
S+	0.4378	0.4263	−0.0115
S−	0.7854	0.7800	−0.0054
E+	1.2875	1.3006	0.0131
E−	1.9764	1.9458	−0.0306

Source

Table 1: True bandpass-filter centre frequencies ω^*_{true} extracted from the BMI weight file, compared with empirically recovered values ω^*_{rec} from the open-loop frequency sweep (Figure 7). $\Delta\omega = \omega^*_{\text{rec}} - \omega^*_{\text{true}}$. Muscles are ordered by ascending filter frequency: shoulder flexion (S+), shoulder extension (S−), elbow flexion (E+), elbow extension (E−). Source

We observe that amplitude decreases with frequency, making it harder to drive muscles that require higher frequencies. This may be due to the low-pass action of the `Brain`. We also observe cross-talk for elbow movements but not shoulder movements: the shoulder can be moved independently of the elbow, but the elbow cannot be moved indepedently of the shoulder.

We then ask whether neural observations carry a muscle-specific spatial code: if different neurons respond preferentially at each selection frequency, the control target could be weighted to favour one muscle over another. For each selection frequency we record the full $n_y = 16$ D amplitude spectrum and normalise the per-neuron profile by its own neuron-mean (Figure 8).

Muscle identity is therefore encoded temporally by the oscillation frequency of x_1, not spatially by which neurons are active (Figure 4). No differential weighting of neurones in the spectral control target offers muscle selectivity beyond what the BMI's bandpass filters already provide. This justifies controlling the population mean at the spectral target.

Control method

The core pipeline is a 3 step process. First is band-identification, an open-loop sinusoidal sweep over both brain input channels, with cumulative shoulder and elbow displacement (inferred via inverse kinematics) recorded per frequency. The four bands are identified, which correspond to S+, S−, E+, and E−. The next step is gain calibration, where a short-closed loop LQI trial at each band frequency targets a sinusoidal spectral setpoint, and mean joint angle per unit amplitude forms a 2×4 gain matrix where rows are joints, and columns are bands. The third is the main control loop, where inverse kinematics converts cartesian target to a joint-angle goals, band amplitudes are solved from the gain matrix and an LQI controller optionally closes the loop on the population mean.

The elbow to shoulder cross-talk motivates a control strategy that first achieves the correct elbow angle, then reduces control gain over the elbow, and focuses on using S+ and S− to correct the shoulder angle. In addition, during the initial correction angle phase, we should alternate between correcting the elbow angle and correcting the shoulder angle, as attempting to move the elbow moves the shoulder. To mitigate cross-talk further, we drive the shoulder in the opposite direction of the expected cross-talk.

We always begin with targeting elbow error first. Each elbow/shoulder phase has a minimum length and maximum length. In addition, when controlling the elbow, there is a maximum shoulder drift, such that if upon moving the elbow the shoulder drifts too much, we switch to controlling the shoulder. We also provide per-band gain multipliers applied to the gain matrix values before the solve, to further tune individual band aggressiveness.

Hyperparameter optimisation

With so many model parameters defined by the complex sequential controller design, we turn to hyperparameter optimisation to search the parameter space for an optimal solution. We use Tree-structured Parzen Estimator (TPE) via the `optuna` library to run two phases: first, a large set of Bayesian Optimisation (BO) trials; then the top trials are re-evaluted on a full target grid, and the best is selected. We define an objective score:

$$\mathcal{S} = \overline{d}_{\text{final}} - 20\,\overline{f}_{\text{sector}} - 10\,\overline{p}_{\text{reached}} + 50\,\overline{\max(0,\ \theta_{\text{rot}} - 2\pi)}$$

This directly penalises final distance, inefficient trajectories, and rewards successful reach rate to create a more interpretable optimisation process. Figure 9 shows how the optimisation process did not reduce final distance or the time to threshold, but instead made the trajectory more sensible, increasing the sector fraction and reducing the rotation.

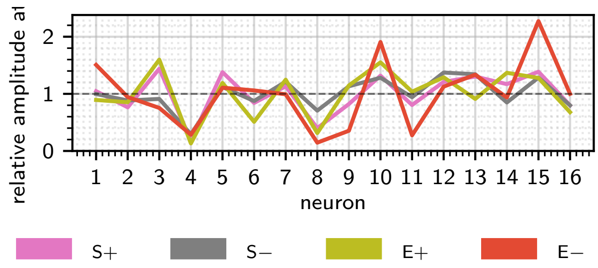

Figure 8: Neural correlates of muscle selection. For each of the four selection frequencies identified in Figure 7, a sustained open-loop sinusoidal trial ($u_0 = 0.5 + 0.4\sin(2\pi f^* t)$, $u_1 = 0.5$) recorded $n_y = 16$ D neural observations over 1024 steps following 500 burn-in steps. Each line shows the amplitude at ω^*_{sel} per neuron, normalised by the muscle's own neuron-mean; the dashed line marks the mean ($= 1$). Profiles that collapse onto each other confirm that every muscle recruits the same relative neural activation pattern, so the 16 D $\to x_1$ projection is not muscle-specific. Source

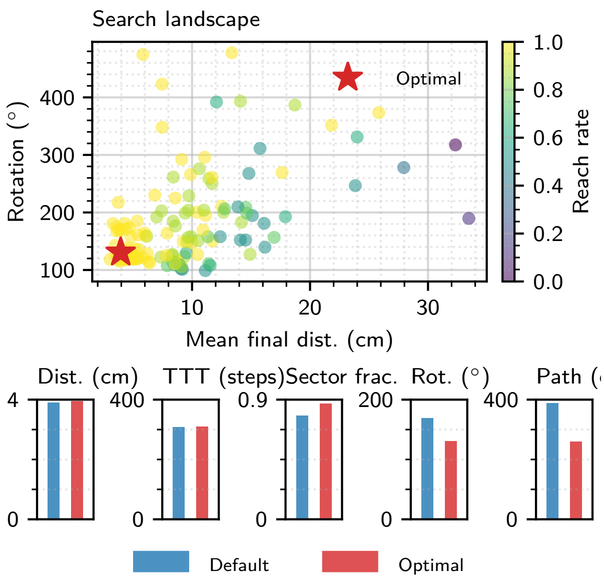

Figure 9: Bayesian optimisation (TPE sampler, 150 trials) over eight controller parameters (`seq_blend_alpha`, `el_hold_alpha`, `sh_vel_damp`, `min_phase_steps`, `max_phase_steps`, `use_lqi`, and the S± and E± band scales independently). The composite score is $\overline{d}_{\text{final}} - 20\overline{f}_{\text{sector}} - 10\overline{p}_{\text{reached}} + 50\max(0,\ \theta_{\text{rot}} - 2\pi)$ (lower is better; the final term heavily penalises hand trajectories that exceed one full revolution, discouraging looping behaviour). Left: per-trial scores (faint points) and running best (line). Centre: search landscape; each point is one BO trial at (mean final distance, sector fraction), coloured by reach rate; the optimal Phase-B config is starred. Right: four per-metric comparisons of the default vs. optimal config on the full 24-target grid. Optimal params: `seq_blend_alpha`=0.137, `el_hold_alpha`=0.622, `sh_vel_damp`=30.3, `min_phase_steps`=30, `max_phase_steps`=78, `use_lqi`=False, `band_scale`=[1.5, 1.5, 0.24, 0.24]. Final distance: 3.9 cm (default) vs. 4.0 cm (optimal); sector fraction: 0.782 vs. 0.871; mean rotational distance: 169° (default) vs. 131° (optimal). Source

Results

The most optimal controller's metrics are summarised in Table 2, its losses over time are plotted in Figure 10, an example trajectory is illustrated in Figure 11, and a heatmap of targets and the final distances to them is drawn in Figure 12.

Metric	Default (mean $\pm$ SD)	Optimal (mean $\pm$ SD)
Final dist. (cm)	3.90 ± 2.76	3.95 ± 1.42
Reach rate	1.000 ± 0.000	1.000 ± 0.000
TTT (steps)	308 ± 155	309 ± 147
Sector frac.	0.782 ± 0.126	0.871 ± 0.082
Rot. dist. ($^\circ$)	169.3 ± 121.8	130.6 ± 65.5
Path length (cm)	389.4 ± 211.3	259.6 ± 68.4

Source

Table 2: Per-metric comparison of the default and Bayesian-optimised Phase-B controllers on the full 24-target grid (mean $\pm$ SD across targets). TTT is the time to threshold in simulation steps; sector fraction measures the proportion of the trajectory spent in the sector pointing toward the target; rotational distance is the total angular sweep of the hand about the target. The optimiser leaves final distance, reach rate, and TTT unchanged while reducing rotational distance and path length and raising sector fraction (Figure 9). Source

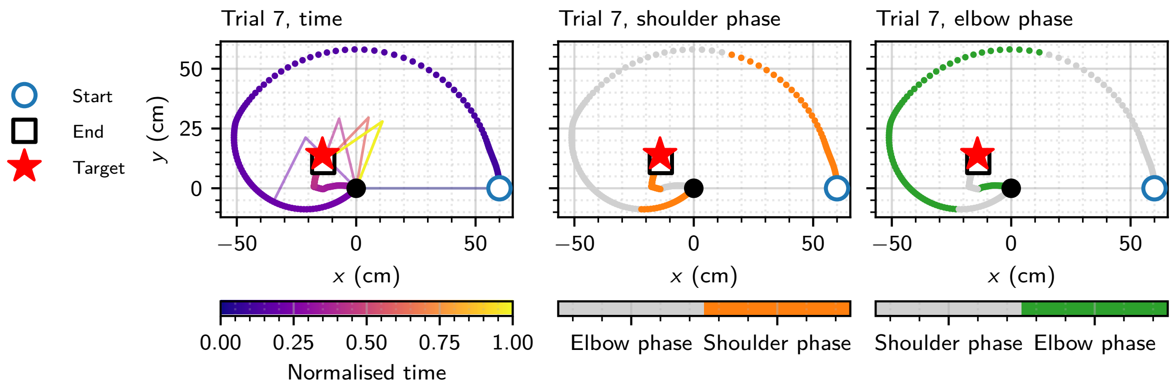

Figure 11: Hand trajectories for trial 7 of the sequential open-loop controller ($T = 1000$ steps, seed 0). Left: path coloured by normalised time (plasma colourmap); 6 arm-posture snapshots are overlaid at evenly spaced steps. Centre: path coloured by the active control phase, orange indicating the shoulder-correction phase and grey the elbow-correction phase. Right: same trajectory with elbow-phase active shown in green. Phase switches: 7. Final distance to target: 3.2 cm. Source

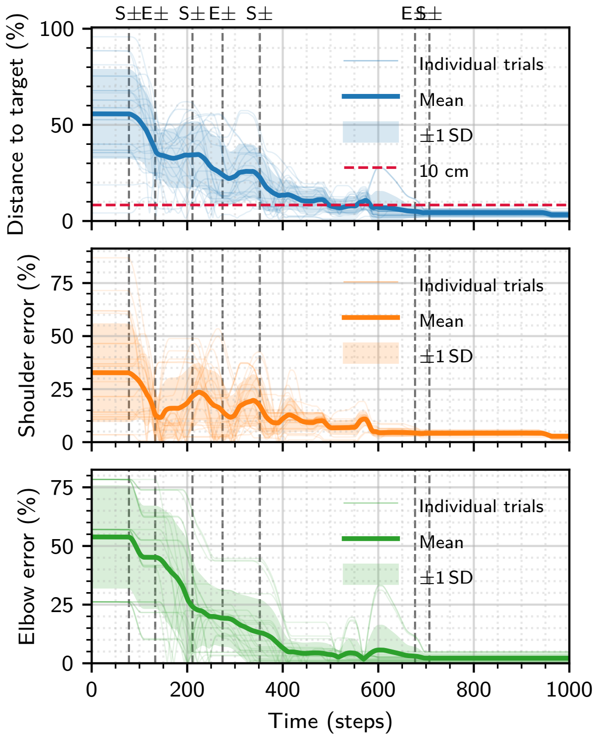

Figure 10: Distance and joint-error time series for the sequential open-loop controller evaluated on 24 targets arranged on a 3×8 polar grid ($r \in [20, 55]$ cm, $T = 1000$ steps, seed 0). Top: distance to target as a percentage of the maximum reachable distance (120 cm); dashed line marks the 10 cm success threshold. Centre: shoulder joint error as a percentage of the maximum wrapped angular error (180°). Bottom: elbow joint error on the same scale. Faint traces show individual trials; solid line and shading show mean ± 1 SD. Vertical dashed lines mark phase transitions in the demo trial (trial 7), labelled by the phase activated: S± (shoulder correction) or E± (elbow correction). 24/24 trials reached within 10 cm (mean time = 309 steps); mean final distance = 4.0 cm. Source

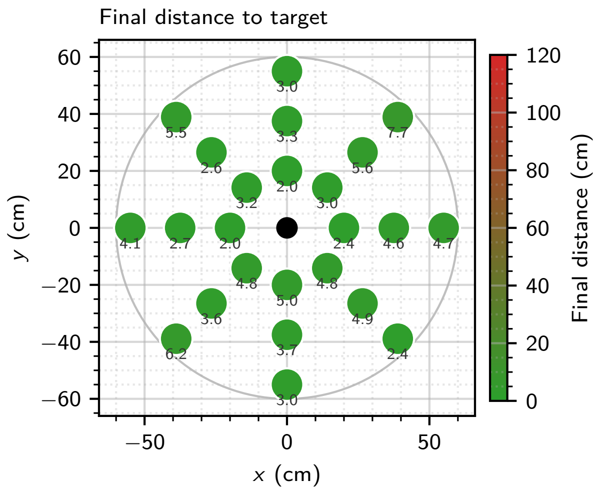

Figure 12: Final distance to target (cm) at each grid location for the sequential open-loop controller. 24 targets on a 3×8 polar grid ($r \in [20, 55]$ cm; $T = 1000$ steps, seed 0). Dot colour encodes final Euclidean distance: green (0 cm) to red (>20 cm). The faint grey circle marks the arm reach boundary ($2l$ where l is the link length). Mean final distance = 4.0 cm (SD = 1.4 cm). Source

Sensitivity to model parameters

We carry out ablation analysis, varying one parameter of the controller in isolation and holding everything else fixed to measure how sensitive performance is to that parameter. Hyperparameters under analysis include: the elbow band scale, how much the elbow gain is turned down relative to the shoulder gain Figure 13; the blend factor, how much we compensate for cross-talk by driving the shoulder in the opposite direction Figure 14; closed-loop vs open-loop control Figure 15; and whether we constrain the elbow to a maximum of 180° or not Figure 16. We filter the metrics using a one-way Analysis of Variance (ANOVA), run independently for each metric across all conditions, with a null hypothesis that all group means are equal. A metric is plotted only if the ANOVA p-value satisfies $p \leq 0.05$.

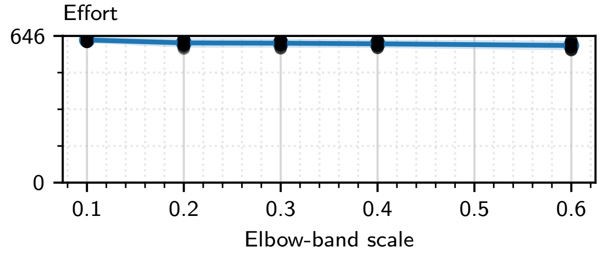

Figure 13: Ablation over the E$\pm$ band-scale gain $\in \{0.1, 0.2, 0.3, 0.4, 0.6\}$ (S$\pm$ fixed at 1.0) on 24 targets across $N = 10$ Brain seeds ($T = 1000$ steps). Each panel shows a different metric (top-left: final distance; top-right: sector fraction; bottom-left: time-to-threshold; bottom-right: effort). Line and shading show mean $\pm$ 1 SD across seeds; dots show individual seed means. The elbow-band scale modulates the gain-matrix rows corresponding to E$\pm$ bands, controlling how strongly the elbow correction signal drives the brain inputs. Source

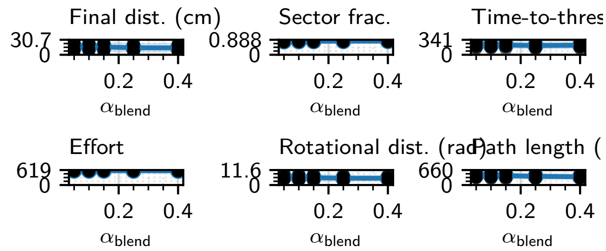

Figure 14: Ablation over the idle-joint sequential blend parameter $\alpha_{\text{blend}} \in \{0.05, 0.1, 0.15, 0.25, 0.4\}$ on 24 targets across $N = 10$ Brain seeds ($T = 1000$ steps). Each panel shows a different metric (top-left: final distance; top-right: sector fraction; bottom-left: time-to-threshold; bottom-right: effort). Line and shading show mean $\pm$ 1 SD across seeds; dots show individual seed means. α_{blend} attenuates the idle joint during each sequential phase; lower values suppress cross-talk at the cost of slower convergence. Source

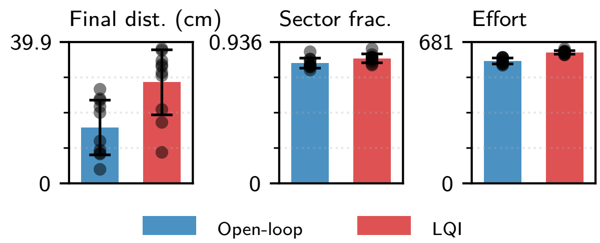

Figure 15: Ablation comparing open-loop (sinusoidal drive) and closed-loop LQI control modes on 24 targets across $N = 10$ Brain seeds ($T = 1000$ steps). Bars show mean across seeds; error bars and dots show ś1 SD and individual seed means, respectively. Four metrics are shown: final distance to target, sector fraction, time-to-threshold, and control effort. Open-loop final distance: 15.8 ± 7.7 cm; LQI: 28.5 ± 9.2 cm. Source

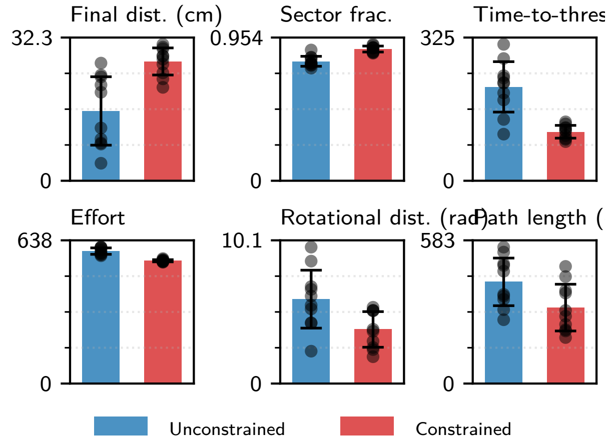

Figure 16: Ablation comparing unconstrained and anatomically constrained elbow ($\theta_{\text{el}} \in [0, \pi]$) on 24 targets across $N = 10$ Brain seeds ($T = 1000$ steps). Bars show mean across seeds; error bars and dots show ś1 SD and individual seed means. Unconstrained final distance: 15.8 ± 7.7 cm (elbow error 29.0°); constrained: 26.9 ± 3.1 cm (elbow error 5.3°). Source